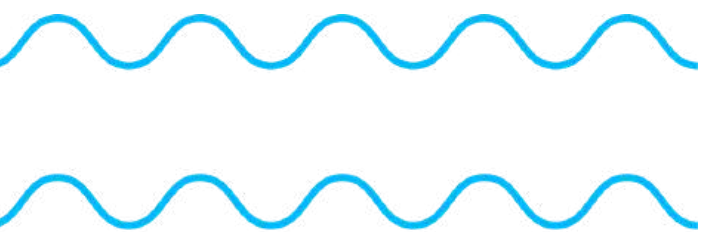
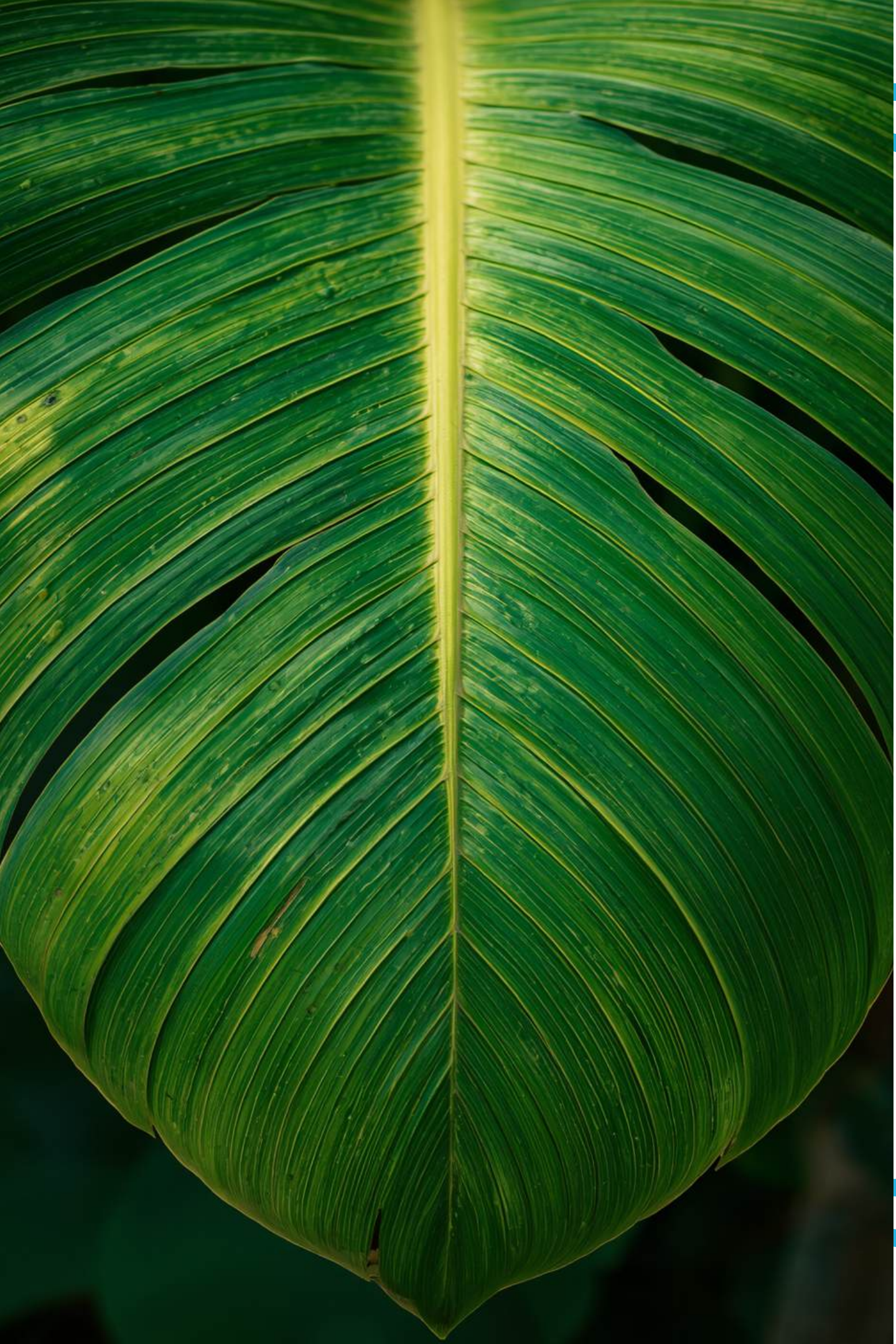


Biomimicry — Nature's Engineering





What is Biomimicry?

Biomimicry means copying nature's ideas to solve human problems. Scientists and engineers look closely at plants and animals to find smart solutions!

💭 Can you think of something in nature that helps us?

Kingfisher Beak Inspires Silent Trains

The kingfisher bird dives into water at high speed without making a big splash. Its pointed beak cuts through air and water smoothly. Engineers copied this clever shape to design the nose of bullet trains!

Kingfisher's beak helps it dive quietly into water. Engineers copied this shape to make trains run silently and much faster!

Pause and Think: Why do you think quiet trains are better? How does less noise help people living near train tracks?



Lotus Leaves Keep Themselves Clean

Lotus leaves have tiny bumps on their surface that make water roll off and carry dirt away. This amazing ability is called the "Lotus Effect" — and scientists have copied it to create self-cleaning surfaces!

Tiny bumps on the lotus leaf push water into droplets. As the droplets roll away, they pick up dust and dirt — keeping the leaf perfectly clean!

💡 Pause and Think: How can we use this idea in our homes or clothes? Think of windows, school uniforms, or rooftops!



Termite Mounds Keep Cool Without AC

Termites build amazing mounds with tunnels and chimneys that let cool air flow through. These mounds stay cool inside even when it is very hot outside — no electricity needed!



Engineers in India and around the world are copying this idea to design buildings that stay cool using natural air flow — saving energy and reducing electricity bills!

 **Pause and Think:**
How would a cool home help us in summer? What problems would it solve for families in India?

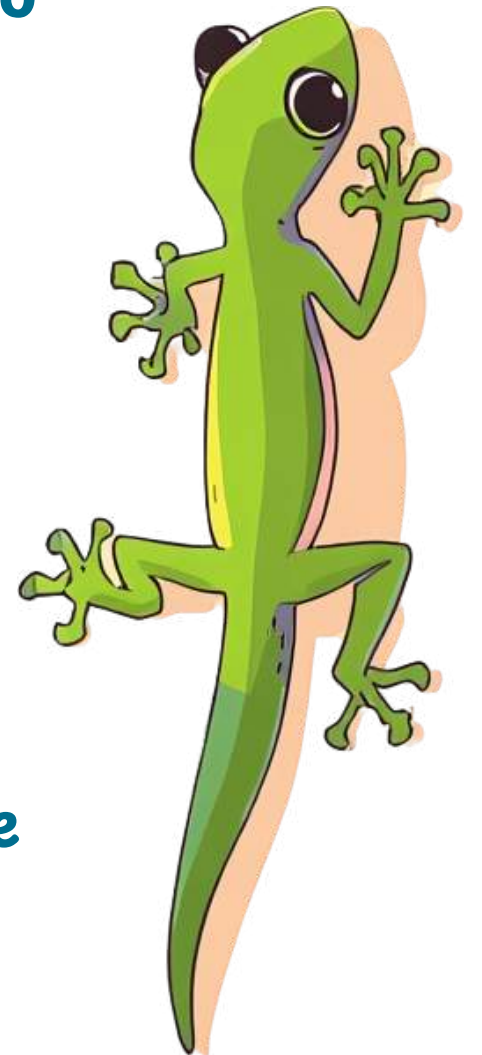
Gecko Feet Stick Without Glue!

Geckos have thousands of tiny hairs on their feet. These hairs grip surfaces so well that geckos can climb walls and even ceilings — without any glue at all!



Scientists studied gecko feet and copied the tiny hair idea to make sticky shoes, climbing gloves, and special tapes that stick and unstick easily!

 **Pause and Think:**
"What else could we stick to walls for? Can you think of a fun invention using gecko-style stickiness?"



Design Like Nature!

Think of a problem around you. Now, imagine how an animal or plant solves it. Draw or describe your own biomimicry idea — there are no wrong answers!

Pick a problem: slipping on wet floors, staying cool in summer, or carrying heavy things.

Find your nature hero: a gecko, lotus leaf, termite, kingfisher — or any plant or animal!

Draw your invention or write how it works. Label the nature idea you borrowed!

Share your design with the class next time — your idea could inspire real science!



Quiz Time!

Question 1: Which bird helped engineers design silent trains?



Think carefully and pick your answer!

(A) Peacock

(B) Kingfisher

(C) Sparrow

(D) Eagle

Quiz Time!

Question 2: What special property does a lotus leaf have?

**think before
you answer!**

(A) Sticky

(B) Self-cleaning

(C) Colourful

(D) Soft



Quiz Time!

Question 3 : How do termite mounds stay cool?

**think before
you answer!**

(A) Air flow

(B) Water inside

(C) Shade

(D) Electricity



Your Nature Challenge!

Look around your home or school. Find one thing in nature that solves a problem. Draw a picture or write about it! Bring your ideas next time to share with the class!

Step 1: Go outside or look around your home or school garden.

Step 2: Find a plant, insect, or animal that does something clever!

Step 3: Draw a picture or write about what you observed.

Step 4: Bring your ideas next class and share with everyone!



Thank You!

Keep watching nature and
inventing new ideas!

